# University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

## FACULTY OF INFORMATION TECHNOLOGY

**BSSE Roadmap for Batch Fall '22, Spring '23**

| Semester - 1 | | | | Semester - 2 | | |
|---|---|---|---|---|---|---|
| **S. #** | **Course Title** | **Cr Hrs** | | **S. #** | **Course Title** | **Cr Hrs** |
| 1 | Introduction to Computing | 3+1 | Year - 1 | 1 | Programming Fundamentals | 3+1 |
| 2 | Basic Electronics | 3 | | 2 | Communication & Presentation Skills | 3 |
| 3 | English Composition & Comprehension | 3 | | 3 | Discrete Structures | 3 |
| 4 | UCP Elective-I | 3 | | 4 | Software Engineering | 3 |
| 5 | Pakistan Studies | 2 | | 5 | Calculus and Analytical Geometry | 3 |
| 6 | Fundamentals of Entrepreneurship | 1 | | 6 | | 3 |
| **Total** | | **16** | | **Total** | | **16** |
| Pre-Calculus (only for Pre-Med students) | | | | Elementary Algebra (only for Pre-Med students) | | |
| **Semester - 3** | | | | **Semester - 4** | | |
| **S. #** | **Course Title** | **Cr Hrs** | | **S. #** | **Course Title** | **Cr Hrs** |
| 1 | Object Oriented Programming | 3+1 | Year - 2 | 1 | Data Structures & Algorithms | 3+1 |
| 2 | Software Requirement Engineering | 3 | | 2 | Software Design & Architecture | 2+1 |
| 3 | Database Systems | 3+1 | | 3 | Probability and Statistics | 3 |
| 4 | Linear Algebra | 3 | | 4 | Web Engineering | 3 |
| 5 | Islamic Studies | 2 | | 5 | Technical & Business Writing **(ENG-III)** | 3 |
| **Total** | | **16** | | **Total** | | **16** |
| **Semester - 5** | | | | **Semester - 6** | | |
| **S. #** | **Course Title** | **Cr Hrs** | | **S. #** | **Course Title** | **Cr Hrs** |
| 1 | Operating Systems | 3+1 | Year - 3 | 1 | Computer Networks | 3+1 |
| 2 | Human Computer Interaction | 3 | | 2 | Software Construction and Development | 2+1 |
| 3 | SE Elective – I | 3 | | 3 | SE Elective – II | 3 |
| 4 | University Elective –II | 3 | | 4 | SE Supporting - II | 3 |
| 5 | SE Supporting - I | 3 | | 5 | SE Elective -III | 3 |
| | | | | | | |
| **Total** | | **16** | | **Total** | | **16** |
| **Semester - 7** | | | | **Semester - 8** | | |
| **S. #** | **Course Title** | **Cr Hrs** | | **S. #** | **Course Title** | **Cr Hrs** |
| 1 | Information Security | 3 | Year - 4 | 1 | Professional Practice | 3 |
| 2 | Software Project Management | 3 | | 2 | Software Quality Engineering | 3 |
| 3 | Software Re-Engineering | 3 | | 3 | SE Supporting -III | 3 |
| 4 | SE Elective – IV | 3 | | 4 | University Elective - III | 3 |
| 5 | SE Elective – V | 3 | | 5 | University Elective - IV | 3 |
| 6 | Final Year Project - I | 3 | | 6 | Final Year Project - II | 3 |
| **Total** | | **18** | | **Total** | | **18** |
| **Total Credit Hours.** | | | | | | **132** |